Abhishek Sarkar, PhD

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Research Scientist, Department of Electrical Engineering
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Research Interests

My research explores fundamental information-theoretic questions for physical systems, spanning both the classical and quantum regimes. I try to provide *simple* answers to questions related to communication, storage and computations over such physical systems, employing the universal language, mathematics.

Education

- **Ph.D. in Electrical Engineering**, IIT Bombay, Mumbai, India **2026**
  - *Dissertation:* Identification via Permutation Channels
  - *Advisor:* Prof. Bikash Kumar Dey
  - *Co-Advisor:* Prof. Sibi Raj B. Pillai
- **M.Tech. in Communication & Signal Processing Engineering**, IIEST Shibpur, India **2018**
- **B.Tech. in Electronics and Communication Engineering**, WBUT, Kolkata, India **2015**

Publications

Peer-Reviewed Journal Publications

1. **A. Sarkar** and B.K. Dey, "Identification Over Permutation Channels" *IEEE Transactions on Information Theory*, vol. 71, no. 9, pp. 6668 - 6691, 2025. [DOI]

Peer-Reviewed Conference Publications

1. **A. Sarkar** and B.K. Dey, "Message Identification Over Noisy Permutation Channels: Deterministic Encoding and Feedback," accepted for presentation in *Proc. IEEE International Symposium on Information Theory (ISIT)*, 2026 (will be held in June, 2026)
2. **A. Sarkar** and B.K. Dey, "Message Identification Over Binary Noisy Permutation Channels," in *Proc. IEEE International Symposium on Information Theory (ISIT)*, Michigan, USA, 2025.
3. **A. Sarkar** and B.K. Dey, "Identification via Binary Uniform Permutation Channel" in *Proc. IEEE Information Theory Workshop (ITW)*, Shenzhen, China, 2024.
4. **A. Sarkar**, B.K. Dey and S.R.B. Pillai "On the Error Exponents for Common Message Broadcasting over DMCs with Feedback" in *Proc. IEEE International Conference on Signal Processing and Communications (SPCOM), Bangalore, India, 2022.*

In Preparation

1. **A. Sarkar** and B.K. Dey, "Reliable Transmission over Weight-constraint Permutation Channels" in preparation.
2. **A. Sarkar**, B.K. Dey and S.R.B. Pillai "On the Rate-limited feedback Communication over AWGN Channels" in preparation.
3. **A. Sarkar** and B.K. Dey, "Identification Over Noisy Permutation Channels" [arXiv]

Work Experience

- **Project Research Scientist**, IIT Bombay, Mumbai, India **Feb, 2026 – Present**
  - We are studying reliable communication over Gaussian permutation channels.
- **Research Scholar**, IIT Bombay, Mumbai, India **July, 2019 – January, 2026**
  - We studied “message identification” (Beyond the classical Shannon paradigm) over permutation channels. We developed a new quantization scheme for an arbitrary probability distribution, which plays a crucial role in the proof the results.
  - We explored “reliable transmission” (the classical Shannon paradigm) of messages over a weight constraint permutation channel and designed a novel concatenated code construction for the direct part.
  - We explored the application of Polar codes over an AWGN channel with rate-limited feedback.
  - Experience in GNU Radio
- **Assistant Professor**, UEM Kolkata, Kolkata, India **July, 2018 – June, 2019**

Poster Presentations

- “Identification Over Noisy Permutation Channels” in *Recent Results Session*, IEEE International Symposium on Information Theory (ISIT), Michigan, USA, 2025.

Others

- **Teaching Assistant**- Course: Digital Communications, Real Analysis, Finite Fields and Information Theory & Coding at IIT Bombay
- Conducted reading groups about information theory and wireless communication at IIT Bombay.